

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? YES

Does R2 show up as an OSPF neighbor on R3? NO

What does this information tell you?

Answer: It informs that R2 has returned R1 to its OSPF adjacency since the passive interface set-up on S0/0/0 of R2 has been removed. But the S0/0/1 interface of R2 (connects to R3) is still passive. Since R3 is unable to establish a direct neighbor relationship with R2, it must pass traffic to R2 LAN sub-optimally via R1. The cumulative cost of 129 is the cumulative cost of S0/0/0 interface cost(64) of R3, S0/0/0 interface cost(64) of R1 and G0/0 interface cost(1) of R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost will be 65, 64 from the R3's outgoing S0/0/1 interface and 1 from the R2's outgoing G0/0 interface.

Is R2 listed as an OSPF neighbor to R3? _YES_

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because the OSPF cost metric must be an integer, any link with a transmission speed of 100 Mb/s or higher will automatically be assigned a cost of 1. Changing the reference-bandwidth to a higher value allows OSPF to accurately differentiate and calculate the routing costs for networks that utilize links faster than 100 Mb/s.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost = **Reference bandwidth / Interface Bandwidth**, the default reference bandwidth was set to 100Mb/s and the bandwidth on R1's S0/0/0 and S0/0/1 interfaces is set to 128 Kb/s.

The cost for both of R1's serial interfaces is $100 \times 1000 / 128 = 781.25$ and rounds down to **781**.

Cost to the 192.168.3.0/24: $781 + 1$ from the R3 G0/0 = **782**.

Cost to the 192.168.23.0/30 : R1 S0/0/1 and R3 S0/0/1 $781 + 64$ from default T1 bandwidth = **845**

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, since we calculated the new serial link cost is **781**, consider that to reach the 192.168.23.0 network from R1, the traffic must exit R1's serial interface and then exit either R2's or R3's connecting serial interface. Adding those two outbound interface costs will get the cost $781 \times 2 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will always prefer the path with the lowest accumulated cost. By issuing the "ip ospf cost 1565" command to manually force R1-S0/0/1 cost to 1565 + 1(R3-G0/0) = 1566. And Now the lowest cost path become the indirect path (via R2) which from R1-S0/0/0 (cost 781) to R2-S0/0/1 (cost 781) and finally R3-G0/0 (cost 1). The total cost = $781 * 2 + 1 = 1563$. Because ~~1565~~ $1565 > 1563$, Choose indirect path .

1566

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because the OSPF router ID is used to uniquely identify the router within the OSPF routing domain. Controlling the assignment (such as by using a loopback interface or the "router-id" command) ensures stability.

If a router ID is tied to a physical interface and that interface goes down, the router ID changes, which can force the OSPF process to recalculate and disrupt routing.

2. Why is the DR/BDR election process not a concern in this lab?

Because in the lab, the connections between routers R1, R2, and R3 are all configured over serial links and OSPF treats serial links as Point-to-Point network types, which do not require a Designated Router (DR) or Backup Designated Router (BDR). DR/BDR elections only occur on multi-access networks

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive prevents the router from sending OSPF routing updates (Hello packets) out of that specific interface. It can reduce unnecessary bandwidth consumption and process overhead on LANs where no other OSPF routers exist.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF's use of Areas creates a hierarchical design. By dividing a large OSPF domain into a central Backbone Area (Area 0) and peripheral areas, it is able to contain routing updates and topology recalculations to specific zones. This keeps routes small, minimizes router memory/CPU usage, and allows the network to scale massively.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The **EIGRP** route will be chosen. When a router learns about the exact same destination network from different routing protocols, it does not compare their metrics (since they use completely different calculations). Instead, it relies on the **Administrative Distance (AD)**, which is the router's measure of trustworthiness for the protocol. The router always chooses the lowest AD.

Since EIGRP has an AD of 90, OSPF has a default AD of 110, and RIP has an AD of 120. Therefore, EIGRP wins strictly based on Administrative Distance.